

HARDWARE DOCUMENTATION

Smart Level Crossing Guardian

An intelligent, sensor-fused safety system for unmanned railway crossings

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Contents

1. System Overview

Purpose. The Smart Level Crossing Guardian is an electronic control system that automatically detects an approaching train at an unmanned railway crossing and lowers a barrier, while warning road users with visual and audible alerts. It is designed to replace timer-based or manually operated gates, which suffer from delayed activation, false alarms, and an inability to detect obstacles on the track.

Operating principle. The system follows a four-stage loop: it senses the track with a LIDAR distance sensor, decides whether a genuine train is approaching by fusing the LIDAR reading with a rail-vibration signal, acts by driving a servo barrier and alert devices, and records each event for analytics. Requiring agreement between two independent sensors is the system’s key safety feature — it rejects false alarms caused by animals, stray objects, or sensor noise that a single-sensor gate would act on.

This document covers. The electronic hardware design: the bill of materials, the schematic, the ESP32 pin assignments, the power architecture, the PCB layout and its design decisions, assembly notes, and a verification checklist.

Attribution. This hardware extends the LIDAR-based crossing-automation concept described by Kulkarni, Patange & Pawar (IJISRT, 2024). The sensor-fusion logic, the PCB design, and the analytics layer described here are this project’s own contributions.

2. Functional Description

Functionally the board is organised into four groups, which map directly onto the connectors and components on the PCB:

Group	Components	Function
Power	PowerInput connector, AMS1117-3.3 regulator, C1/C2/C3	Accepts 5V input and derives a clean, regulated 3.3V rail for the controller
Control	ESP32-WROOM-32 module, R3 (EN pull-up)	Runs the firmware: reads sensors, runs the decision logic, drives outputs
Sensing	J1 (LIDAR), and the vibration input	Provides distance and rail-vibration signals used for train detection
Output	J2 (servo), J4 (LEDs + buzzer), D1, D2, R1, R2, BZ1	Drives the barrier, status lights, and audible alert

3. Bill of Materials

All components are through-hole except the AMS1117 regulator (SOT-223), chosen for ease of hand-assembly on a prototype board.

Ref	Component	Value / Part	Pkg	Qty	Purpose
U1	ESP32-WROOM-32	ESP32 module	Module	1	Main controller / decision brain
U2	LDO regulator	AMS1117-3.3	SOT-223	1	5V → 3.3V regulation for ESP32
J3	Power input	Conn 1x02	2.54mm	1	5V supply input (+5V, GND)
J1	LIDAR connector	Conn 1x04	2.54mm	1	TF-Luna: VCC, RX, TX, GND
J2	Servo connector	Conn 1x03	2.54mm	1	SG90 servo: V+, signal, GND
J4	LED/buzzer conn	Conn 1x03	2.54mm	1	External alert devices
C1	Capacitor	10 μ F	Radial	1	Regulator input decoupling
C2	Capacitor	22 μ F	Radial	1	Regulator output bulk decoupling
C3	Capacitor	100 nF	Disc	1	ESP32 high-frequency decoupling
R1, R2	Resistor	220 Ω	Axial	2	LED current limiting
R3	Resistor	10 k Ω	Axial	1	ESP32 EN pull-up
D1	LED	Green	5mm	1	Gate-open indicator
D2	LED	Red	5mm	1	Gate-closed indicator
BZ1	Buzzer	Active	—	1	Audible alert

External components (connect via the headers, not on the board): TF-Luna LIDAR sensor (J1), SG90 servo motor (J2), and the 5V power source feeding the power input. A rail-mounted vibration sensor provides the second detection signal used in sensor fusion.

4. Schematic

The complete circuit schematic is shown below. Power enters at the lower-left, is regulated to 3.3V by U2, and distributed to the ESP32 and sensors. The ESP32's GPIO outputs drive the servo, LEDs (through current-limiting resistors), and buzzer.

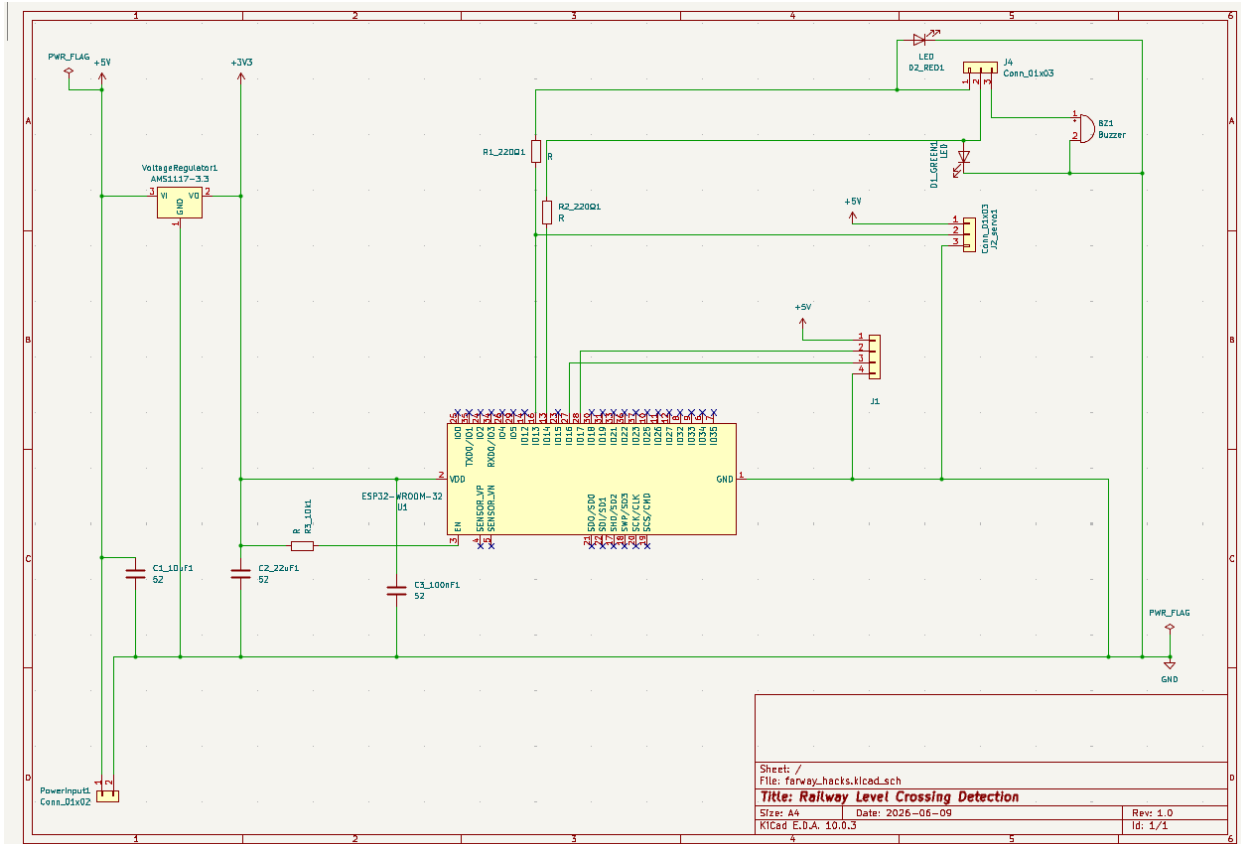


Figure 1 — Complete circuit schematic (KiCad). Verified with a clean Electrical Rules Check (ERC).

Verification: the schematic passed the KiCad Electrical Rules Check (ERC) with zero violations. Power rails were confirmed net-by-net using the highlight tool — +5V reaches only the regulator input and the LIDAR/servo VCC pins, +3V3 reaches only the regulator output, ESP32 VDD, the EN pull-up and decoupling caps, and all grounds tie to a common net.

5. ESP32 Pin Assignments

The firmware and wiring agree on the following GPIO assignments. Unused ESP32 pins are marked no-connect in the schematic.

ESP32 Pin	Direction	Connects to	Function
VDD (pin 2)	Power	+3V3 rail	3.3V supply from regulator
GND (multiple)	Power	GND plane	Common ground (incl. thermal pad)
EN (pin 3)	Input	+3V3 via R3 (10k)	Enable / reset pull-up
GPIO16	Input	J1 LIDAR TX	Receives LIDAR data (UART RX)
GPIO17	Output	J1 LIDAR RX	Transmits to LIDAR (UART TX)
GPIO13	Output	J2 servo signal	Barrier servo PWM
GPIO14	Output	R1 → D2 (red)	Gate-closed LED
GPIO12	Output	R2 → D1 (green)	Gate-open LED
GPIO25	Output	BZ1 buzzer	Audible alert
GPIO4	Input	Vibration sensor	Rail-vibration signal (fusion input)

UART cross-over note: the LIDAR's TX connects to the ESP32's RX (GPIO16) and the LIDAR's RX connects to the ESP32's TX (GPIO17). Transmit must always connect to receive — never TX-to-TX.

6. Power Architecture

A single 5V supply powers the board. The **AMS1117-3.3** low-dropout regulator (U2) converts this to a regulated 3.3V rail for the ESP32, which must not be powered above 3.3V. Devices that require 5V — the LIDAR and the servo — are fed directly from the 5V rail, while the ESP32 and its logic run on 3.3V.

- **5V rail:** PowerInput → regulator VIN, LIDAR VCC (J1), servo V+ (J2).
- **3.3V rail:** regulator VOUT → ESP32 VDD, EN pull-up (R3), decoupling caps (C2, C3).
- **Decoupling:** C1 (10μF) on the regulator input, C2 (22μF) on the output for bulk stability, and C3 (100nF) placed close to the ESP32 VDD pin to suppress high-frequency switching noise.

Design decision — wider power traces. The +5V, +3V3 and GND nets are routed on a dedicated power net class at 0.5mm, twice the 0.25mm signal width. This handles the servo's sudden current draw with minimal voltage drop and heating — a real current-carrying-capacity consideration, not a cosmetic choice.

7. PCB Layout

The board is a two-layer design: signal and power traces on the front copper layer, with a continuous ground pour on the back layer acting as a ground plane.

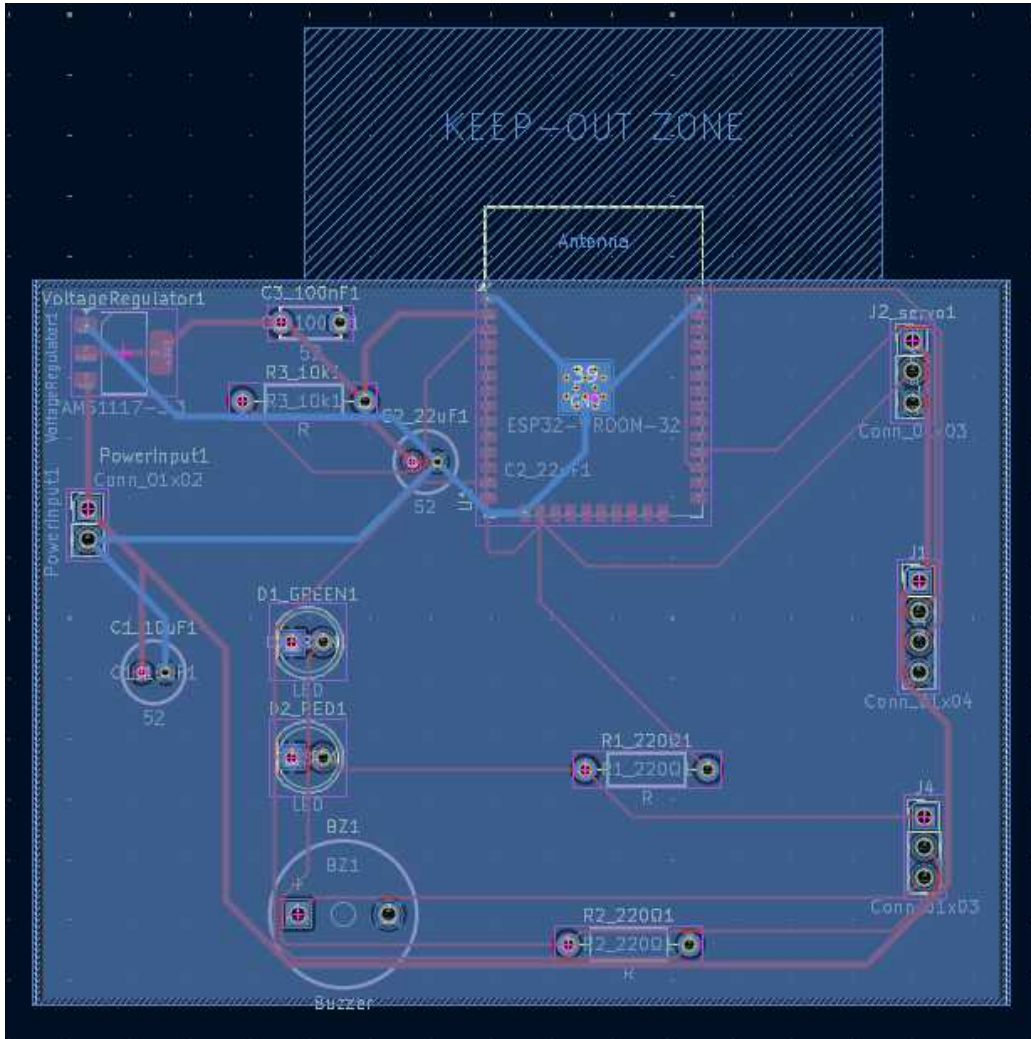


Figure 2 — PCB layout (KiCad). Ground pour on the back layer; the ESP32 antenna keep-out zone faces the board edge.

7.1 Layout decisions

- **Ground plane.** A full copper pour on the back layer (B.Cu) connects every ground pad through the shortest possible return path, reducing noise from the digital ESP32 and the switching servo, and providing shielding.
- **Antenna keep-out.** The ESP32 module is placed so its antenna keep-out zone overhangs the board edge — no copper or components sit beneath the antenna, preserving wireless performance.
- **Decoupling placement.** C3 sits immediately adjacent to the ESP32 VDD pin so its noise-suppression is effective; trace inductance to a distant cap would reduce it.
- **Functional zoning.** Power components are grouped at one corner, connectors line the edges for cable access, and output devices (LEDs, buzzer) are clustered together — minimising trace crossings.

- **Thermal pad grounding.** The ESP32 central thermal pad is tied to the ground net so the module is properly grounded and heat-sunk.

7.2 Design rule verification

The layout passed the KiCad Design Rules Check (DRC) with all electrical connections routed (zero unconnected items). The minimum hole size was set to accommodate the ESP32 footprint's thermal vias. Remaining advisories were cosmetic (silkscreen clipping at the board edge) and do not affect function.

8. Assembly Notes

Recommended hand-assembly order, from lowest to tallest components:

1. Solder the AMS1117 regulator (U2) first — the SOT-223 has a large tab pad that is easy to reach before other parts are in the way.
2. Solder the resistors (R1, R2, R3) and the disc capacitor (C3).
3. Solder the electrolytic capacitors (C1, C2), observing polarity.
4. Solder the LEDs (D1 green, D2 red), observing polarity (long leg = anode).
5. Solder the ESP32 module (U1), then the pin-header connectors (J1–J4) and the buzzer (BZ1).

Polarity-sensitive parts: electrolytic capacitors (C1, C2), both LEDs, and the active buzzer must be installed the correct way round. The regulator and ESP32 are keyed by footprint.

9. Power-On Test & Verification

After assembly, verify in this order before connecting the LIDAR and servo:

- **Continuity / shorts:** with power off, confirm there is no short between +5V and GND, or +3V3 and GND.
- **Regulator output:** apply 5V and measure the +3V3 rail — it should read 3.3V before the ESP32 draws load.
- **Controller boot:** confirm the ESP32 powers up (green LED on, gate open) on first boot.
- **Output test:** with firmware loaded, simulate detection — the servo should drive the barrier, the red LED light, and the buzzer sound.
- **Fusion test:** confirm the gate closes only when both the distance condition and the vibration signal are present — the deciding factor that rejects false alarms.

Pre-hardware validation. Because physical components were still in transit at submission, the firmware and decision logic were validated in a circuit simulation (ESP32 with an ultrasonic sensor standing in for the LIDAR and a button for the vibration sensor). Both return the same signal types as the production sensors, so the verified control logic transfers directly to the assembled board.